

Effect of dietary protein to energy ratio on performance of Nile tilapia and food web enhancement in semi-intensive pond aquaculture

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ABSTRACT

When fish have only access to formulated feed, the optimal dietary protein to energy ratio (P:E) for tilapia ranges between 18 and 23 g.MJ⁻¹. In pond culture, where natural foods complement administered feed, increasing the carbon:nitrogen (C:N) ratio stimulates the natural food productivity. This study assessed if lowering the dietary P:E ratio (and thus increasing the C:N ratio of the feed input in the pond) below the optimal P:E ratio affects fish productivity, food web dynamics and nitrogen balances in semi-intensively managed tilapia ponds. Twelve ponds, each divided into three equally-sized compartments, were assigned to test the effect of two diets, which differed in P:E ratio (19 vs. 14 g.MJ⁻¹). Three feeding levels (no, “low” and “high”) were nested in each pond in a split plot design. Initial fish biomass was 1166 (± 16) g.compartment⁻¹ and the experiment lasted 60 days. Decreasing P:E ratio enhanced tilapia production and specific growth rate ($P < .05$; 1195 vs. 985 g.compartment⁻¹ and 1.76 vs 1.55%.d⁻¹). Body composition of tilapia was unaffected by diet and feeding level. Despite the difference in performance, final fat content was 5% of body weight and unaffected by treatments. Averaged over both diets, survival and feed conversion ratio increased with increasing feeding level ($P < .001$). Diet composition did not alter measured water quality, and abundance and diversity of the related parameters of the food web. The total amount of N accumulated in the pond⁻¹ was lower with the low P:E ratio diet (i.e., low protein diet). The data on N gain and N balance at the pond level suggest that the food web productivity was stimulated by reducing the dietary P:E ratio below the reported optimal levels in the literature. It is hypothesized that the optimal dietary P:E ratio is dependent on the culture intensity (extensive, semi-extensive or intensive pond culture).

1. Introduction

Nutritional studies that formed the basis of National Research Council (NRC) recommendation for dietary requirements in fish (NRC, 1993, 2011), were mostly done in absence of the natural food. In outdoor ponds, which is the most commonly used farming system in Asia (FAO, 2016), fish have access to natural foods besides the formulated feed (Porchas-cornejo et al., 2012; Pucher and Focken, 2017; Rahman et al., 2008; Roy et al., 2012). Therefore, in ponds, not all nutrients required by the culture species need to be provided in the formulated diet, as is the case with cage or flow-through tanks (Verdegem, 2013).

The protein to energy (P:E) ratio is one of the important determinants for quality feed formulation. This P:E ratio is often expressed in term of digestible protein over digestible energy (DP:DE). The optimum DP:DE ratio for common aquaculture species ranges between 18 and

23 g.MJ⁻¹ (Fernandes et al., 2016; Helland et al., 2010; Lanari et al., 1995; Lanari and Agaro, 2002; NRC, 1993). Since in most formulated diets, protein and energy digestibility are similar, the optimal range in DP:DE is similar to the P:E ratio. Recently, Haidar et al. (2018) suggested that the optimal P:E ratio for tilapia is probably below 16 g.MJ⁻¹ based on the observation that fish performance linearly increased with decreasing P:E ratio. Haidar et al. (2018) did not test diets with a P:E ratio below 16 g.MJ⁻¹, but recommended further experiments to determine the optimal P:E ratio.

By increasing the carbon or energy availability in the pond, production can be increased. The latter is mostly done by supplying carbohydrates (not via the fish feed), raising the C:N ratio of nutrient inputs to 15–20 (Asaduzzaman et al., 2010a, 2008; Hari et al., 2006, 2004). When the C:N ratio of the nutrient input raises above 10, heterotrophic bacteria become dominant (Boyd, 1996; Lancelot and Billen,

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1985), contributing substantial amounts of bacterial biomass to the food web. Organic, but also inorganic nitrogen are taken up by heterotrophic bacteria, thus keeping ammonia and nitrite levels in the pond low (Avnimelech, 1999; Hari et al., 2006; Hari et al., 2004)). Heterotrophic bacteria, are a protein source, stimulating the food web and the production of fish grazing on natural foods. Asaduzzaman et al. (2008) found a higher concentration of natural foods in the water column and the benthos, as well as a higher fish production in ponds with a C:N ratio of 15 or higher. The P:E ratio of the combination of diet and carbohydrate input (C:N ratio 20) used by Asaduzzaman et al. (2008) was 9.5 g.MJ⁻¹, assuming protein, fat and carbohydrate contain respectively 23.6, 39.5 and 17.2 kJ.g⁻¹ energy (NRC, 2011). This P:E ratio is 50% of the NRC recommended P:E ratio of 18–23 g.MJ⁻¹.

In pond studies, the P:E ratio was adjusted by application of carbohydrate directly into pond and not via the formulated diet (Asaduzzaman et al., 2010b, 2008; Dauda et al., 2018; Hari et al., 2006, 2004; Kidd et al., 2011; Xu et al., 2018). In this approach feeding and carbohydrate addition are often not synchronized, for instance when feeding 3 times per day, while applying the carbohydrate once in the morning (Asaduzzaman et al., 2010a). In addition, the carbohydrates and metabolic wastes resulting from feeding are not equally distributed in the pond. If the P:E ratio is changed by feed formulation, application of carbohydrates separately from the formulated feed is no longer necessary. Other advantages include, homogenous spreading by the fish of faeces and branchiary and urinary wastes over the pond area and introduction of an organic carbon source already exposed to bacteria in the fish gut, thus facilitating decomposition in the pond. However, in the past, P:E ratio optimization in ponds has never been tested by changing the diet composition.

In this study, two experimental diets were formulated with different P:E ratio. The low P:E diet was compared to a regular P:E ratio tilapia diet. The effects of lowering the P:E ratio in the diet on fish production and nitrogen retention and on natural food availability in the pond were assessed. In addition, the effect of feeding level was also investigated. We hypothesized that when fish receive diets with an equal energy content, fish eating the low protein diet will benefit from a more productive food web, allowing them to compensate the lack of protein in the diet by a higher availability of natural foods for fish in the pond.

2. Methods

Two different diets with a contrast in P:E ratio were tested on Nile tilapia (*Oreochromis niloticus*) in 12 outdoor ponds for 60 days. Ponds were divided into three compartments and three different feeding levels were assigned within each diet by pond in a split plot design.

2.1. Diets

Two diets, which differed in P:E ratio, were prepared. Diets were made by steam pelleting. Pellet size was 2.5 mm. The first diet ("High P:E") was formulated to have a P:E ratio of 19 g.MJ⁻¹, which is within the recommended range for tilapia by NRC (1993). This "High P:E" diet had a C:N ratio of 8.8 g.g⁻¹. The second diet ("Low P:E") was formulated to have an increased C:N ratio (11.8 g.g⁻¹). This was achieved by adding wheat and rice bran in the "Low P:E" diet while reducing the amount of ingredient providing protein into the diet. Both diets were identical in energy content. This was confirmed by the chemical analysis of the feed (Table 1).

2.2. Fish, rearing and housing facilities

All male, juvenile Nile tilapia (*O. niloticus*), 14th generation WorldFish GIFT strain were collected from Genetic Hatchery, a GIFT Nile tilapia Multiplication Center in Bangladesh, for this experiment.

Twelve outdoor ponds, each thirty square meter, in a field experimental station were used for this experiment. Each pond was divided

Table 1

Ingredients and analysed chemical composition of the experimental Nile tilapia (*Oreochromis niloticus*) diets differing in P:E ratio.

Diets		
Ingredients (%)	"High P:E" ratio	"Low P:E" ratio
Maize	19	19
Soybean meal	12	6
Wheat bran		15
Wheat flour	20	20
Rice bran		12
Sunflower meal	12	6
Rapeseed meal	12	6
Meat & bone meal	15	8
Fish meal	5	3
Fish oil	2	2
Vitamin & Mineral premix ^a	1	1
Mono calcium phosphate (MCP)	0.7	0.8
DL Methionine	0.3	0.2
Diamol	1	1
Chemical composition		
Dry matter (DM), (g.kg ⁻¹)	908	910
Crude Protein (CP) (g.kg ⁻¹ DM)	313	244
Fat (g.kg ⁻¹ DM)	55	59
Ash (g.kg ⁻¹ DM)	113	86
Phosphorus (g.kg ⁻¹ DM)	15	11
Carbohydrate ^b (g.kg ⁻¹ DM)	519	611
Gross energy (kJ.g ⁻¹ DM)	19	19
P:E ratio (g.MJ ⁻¹)	19	14
C:N ratio ^c (g.g ⁻¹ DM)	8.8	11.8

^a commercial product made by ACI Godrej Agrovet Private Limited.

^b This is calculated value where Carbohydrate = 1000-CP-FAT-Ash.

^c This is calculated C:N ratio considering 16% N content in the protein and 47, 70 and 50% C content in protein, fat and carbohydrate respectively (Van De Waal and Boersma, 2012).

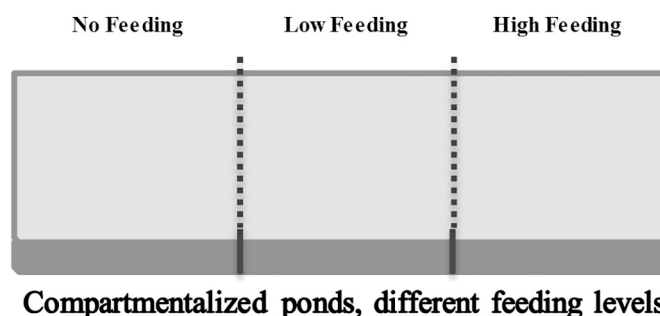


Fig. 1. Sketch of the pond compartments. The dotted line indicates the net & frame, solid bar indicates the concrete block and the grey bottom indicates the mud layer of the pond bottom.

into three equal compartments (Fig. 1). Water column of the pond was divided by bamboo-frame fitted with 1 mm mesh sized nets that allow well mixing of nutrients and dissolved solids within the compartments but prevents passing of the pellets and fish between the compartments. At the bottom, the compartments were separated by concrete block of 105 cm height, of which 75 cm below the soil and 30 cm extended above the pond bottom to prevent the exchange of uneaten feed and benthos between the compartments.

2.3. Experimental procedure

Ponds were dried by pumping out the water. Two hundred fifty g CaCO₃ was applied at the bottom soil of each pond compartment (PC) before water filling. After water filling, 40 g Dolomite (CaMg(CO₃)₂) was spread over the water surface of each compartment. Ten g urea

(CH₄N₂O) and 20 g Triple Super Phosphate (TSP), [Ca(H₂PO₄)₂·H₂O], per pond compartment (Rakocy and McGinty, 1989) were applied 1 week after liming. Fish were stocked in the ponds 5 days after fertilization.

Fish were fed twice a day at 8.00 and 16.00 h. “High” and “low” feeding levels started with 2 & 1% of fish biomass. The feed ration was gradually reduced, reaching 1 & 0.5% at the high and low feeding level compartments, respectively, at the end of the experiment. Feed rations were adjusted based on body weight sampling conducted at day 30 and 45. The amount of feed given of each diet was based on the measured DM content of the diet, thereby ensuring equal amount of feed given of both diets within each feeding level. Current experiment was mimicked with semi-intensive production. The nutrient input to the ponds was kept low to maintain water quality and assure good conditions for mineralization and food web development. So, in this experiment, the “high feeding” level was set at ~50% of the feed input normally applied in semi-intensive ponds in Bangladesh.

Duration of the experiment was 60 days, which was divided into two phases. Phase 1 (Day 1–30) was the time to build the natural food web in the ponds. During this period no measurements were taken. Phase 2 (Day 31–60) was the monitoring period.

2.3.1. Water quality monitoring

From day 31 dissolved oxygen (DO), pH, total dissolved solid (TDS), transparency, temperature and salinity of each pond were measured daily at 6.00, 9.00, 10.00, 12.00, 14.00 and 14.30 h; by using Lutron dissolved oxygen meter model PDO-519, Hanna instruments pocket tester HI98128-phép5, Lutron conductivity meter model PCD-431, Secchi disc, Hanna digital thermometer model HI98501 and Atago refractometer model MASTER-S28 M instruments. NH₄⁺, NO₂ and NO₃ were measured only at day 31, 45 and 60 by colorimetric, acc. to Neßler, with color card and sliding comparator: 108025 | Nitrite Test, 111,117 | Ammonium Test, 110,022 | Nitrite Test; Merck KGaA, Darmstadt, Germany.

2.3.2. Sampling and analysing plankton

Phytoplankton and zooplankton samples were collected at day 31, 45 and 60. Samples were collected between 9.00 and 11.00 h from 3 points, equally spaced on a diagonal line in each pond compartment. At each point 15 L water was passed through the 45 µm mesh plankton net, thus pulling together 45 L of water from each compartment were sampled.

The concentrated samples were preserved in small plastic bottles with 5% buffered formalin. Diversity (group.l⁻¹) and abundance (ind.l⁻¹) estimations of plankton were done using a Sedgewick–Rafter (S-R) cell containing 1000 1-mm³ cells. A 1 ml sample was put in the S-R cell and was left 15 min undisturbed to allow plankton to settle. The plankton in 10 randomly selected cells were identified where possible up to genus level and counted under a binocular microscope (LABOMED America, Inc; Lx 300). Plankton were identified using keys by Ward and Whipple (1959), Prescott (1962), Belcher and Swale (1976), and Bellinger (1992). Plankton abundance was calculated using the following formula:

$N = (P \times C \times 100) / V$. Here, N is the number of plankton cells or units per liter of original water; P, the number of plankton counted in 10 fields; C, the volume of final concentrate of the sample (ml); V = the volume of the pond water sample in liter.

2.3.3. Sampling and analysing benthos

The benthic macroinvertebrate samples were also collected on day 31, 45 and 60 with an Ekman grab (area: 225 cm²). In each pond compartment, bottom mud samples were collected from three different locations, which were then combined into a composite sample. Benthic macroinvertebrates were collected after filtering sediments through 4 different mesh sieve and preserved in a plastic vial containing a 10% buffered formalin. Identification keys used for benthic

macroinvertebrates were Brinkhurst (1971), and Pinder and Reiss (1983). Benthic macroinvertebrates density was calculated using the formula,

$N = Y \times 10,000 / 3A$. Here, N = the number of benthic organisms per square meter; Y = total number of benthic organisms counted in 3 samples; A = Area of the Ekman dredge.

2.3.4. Sampling and analysing bacteria

In order to isolate and quantify the bacterial population, samples from both water and soil sediments were collected at day 31, 45 and 60. All samples were collected from three different locations of each pond compartment in sterile containers (15 ml tube, Falcon, USA), mixed homogeneously before transported back to the Limnological Laboratory of the Environmental Science Discipline of Khulna University, Bangladesh. 1 ml water sample was transferred with a sterile pipette to a test tube containing 9 ml of phosphate buffered saline (PBS) and the tube was shaken thoroughly, while 5 g of each sediment and water samples were weighed and transferred to a sterile conical flask and made up to 50 ml with PBS and the contents were mixed thoroughly to prepare a stock solution. Serial dilution of up to 10–6 for water and 10–8 for sediment were prepared with PBS. Volumes (0.1 ml) of each dilution were spread over the surface of duplicate plates of tryptone soya agar (TSA; Difco, Detroit, MI, USA) with incubation at 30 °C for 24–48 h. Plates with 30–300 colony forming units (CFU) were counted with a Leica Quebec Dark field Colony Counter (Leica, Inc., Buffalo, NY, USA) and expressed as CFU/ml.

2.3.5. Sampling and analysing Chlorophyll a

Water samples were collected from specific locations (Fig. 1) in each compartment of the ponds from the mixed water column. Water samples from three parts of the pond compartments were well mixed and kept into 500 ml bottles. The samples were transferred to the lab within an hour for analysis. There we filtered 250 ml of water through Whatman GF/C filter paper. We then torn the filter paper into 5–6 pieces and inserted them into a 50 ml centrifuge tube. Thereafter, we added 20 ml of methanol into each tube to cover the filter paper pieces in it, shook well and vortex until the filter paper was broken up. Kept them in the freezer overnight. Centrifuged at 3200 rpm for 10 min. Poured off the supernatant into a 1 cm cuvette and Measured the extinction at both 665 nm and 750 nm (zero with methanol). Chlorophyll a was calculated as Chl-a (ug.l⁻¹) = ((Abs[665 nm]–Abs[750 nm]) × A × Vm) / Vf × l. Here, A = absorbance coefficient of chlorophyll-a in methanol (12.63); Vm = volume of methanol used for extinction (ml); Vf = litres of water filtered; and L = path length of cuvette.

2.3.6. Sampling and analysing proximate composition of fish and feed

Initial body composition was determined in 25 fish, which were randomly selected at the start of the experiment. For final body composition, 5 fish were randomly selected per compartment at the end of the experiment. Fish, which were used for body composition analysis, were euthanized by an overdose of a phenoxy-ethanol solution (1.0 ml. l⁻¹) and stored at –20 °C. Before chemical analysis, the sampled fish were cut into small pieces, homogenised by grinding in a mincing machine twice through a 4.5 mm screen grinder and subsequently oven-dried. Chemical analyses were done in triplicate. Dry matter (DM) was determined gravimetrically after drying at 103 °C for 4 and 24 h (h) until constant weight, respectively, for feed and fish samples (ISO 6496, 1983). Crude ash was determined after incineration at 550 °C for 4 h (ISO 5984, 1978). Crude protein (CP) was determined by the Kjeldahl method (ISO 5983, 1979) and calculated by multiplying the measured N content by 6.25. Fat was quantified by petroleum–diethyl ether extraction (ISO 6492, 1999). Before fat analysis, feed samples were hydrolysed by boiling for 1 h with 3 M-HCl. Dietary energy content was measured by direct combustion in an adiabatic bomb calorimeter (IKA-C-7000; IKA analysetechnik, Weikersheim, Germany).

3. Analytical procedures and calculations

3.1. Performance

Biomass gain (g) was calculated as the difference between the biomass stocked and biomass harvested per compartment. The specific growth rate (SGR) was calculated as $SGR = ((\ln(\text{IndBW}_{60}) - \ln(\text{IndBW}_0)) / 60) \times 100$; where IndBW_{60} and IndBW_0 means individual body weight at day 60 and day 0. Feed conversion ratio (FCR) was calculated per compartment using the feed given and weight gain. The survival of fish per compartment was calculated as $(N_f/N_i) \times 100$, where N_f is the final number of fish and N_i the initial number at PC level.

3.2. Nitrogen(N) retention

N Gain in fish was calculated by the difference between N_f and N_i . N Feed was calculated by total feed input per compartment multiplying the N content in feed; N balance was calculated by deducting N gain in fish from total N input from both the feed based on proximate composition and fertilizer (Urea, 46% N). N retained from Food Web was calculated by deducting N retention from feed from the total N gain in fish.

3.3. Sensitivity test to calculate the relative contribution of feed and food web to fish growth

In this study we didn't measure the apparent digestibility. So, the calculation of N retention was based on apparent digestibility coefficient (ADC) 90% for N (Azevedo et al., 2004; Kaushik et al., 1995) and N retention efficiency (RE) of 40% (Azevedo et al., 2004). We assumed same ADC and RE for the N coming from food web. We then made a range of assumptions on the ADC (80%, 85%, 90% & 95%) and RE (30%, 35%, 40% & 45%) of N to evaluate how sensitive the contribution of formulated feed and natural foods to production is to changes in these parameters.

3.4. Statistical analysis

The data were analysed using the IBM SPSS software package version 23. All were analysed for the effect of diet, feeding level and their interaction by two-way repeated measure ANOVA using the procedure general linear model (GLM). When significant interaction found multiple comparisons of means using Tukey's multiple range test were performed.

4. Result

4.1. Fish performance

Fish performance at low P:E diet at all feeding levels was better

Table 2

Effect of dietary protein to energy ratio (P:E) and feeding level on performance of Nile tilapia (*Oreochromis niloticus*).

Per compartment	Units	"Low P:E" diet						"High P:E" diet						P values		
		NF		LF		HF		NF		LF		HF		D	F	D*F
Biomass stocked	g	1193	± 12	1179	± 21	1175	± 16	1155	± 12	1155	± 21	1141	± 16	0.04	0.6	0.9
Biomass harvested	g	1490	± 46	2409	± 133	3233	± 117	1282	± 46	2317	± 133	2805	± 117	0.006	0.000	0.35
Biomass gain	g	297	± 37	1230	± 131	2058	± 125	127	± 37	1162	± 131	1665	± 125	0.014	0.000	0.37
Survival	%	52	± 4	69	± 6	86	± 5	57	± 4	72	± 6	84	± 5	0.64	0.000	0.73
Feed given	g	NA		860	± 20	2025	± 27	NA		880	± 20	2001	± 27	0.93	0.000	0.26
FCR	g.g ⁻¹	NA		0.8	± 0.9	1.0	± 0.9	NA		0.8	± 0.9	1.3	± 0.9	0.09	0.005	0.25
SGR	%d ⁻¹	1.5	± 0	1.8	± 0.07	2.0	± 0.1	1.1	± 0	1.7	± 0.07	1.8	± 0.06	0.02	0.000	0.27

*NF = No Feeding, LF = "Low Feeding" and HF = High Feeding, D = Diet and F = Feeding Level, D*F = Diet and Feeding level interactions.

Table 3

Effect of dietary protein to energy ratio (P:E) and feeding level on final body composition of Nile tilapia (*Oreochromis niloticus*).

g.kg ⁻¹	"Low P:E" diet	"High P:E" diet	Standard Error	P values		
				D	F	D*F
Dry matter	283	287	4.5	0.5	0.6	0.7
Crude protein	167	167	1.4	0.9	0.7	0.4
Crude Fat	52	54	1.6	0.4	0.6	0.7
Ash	39	42	1.2	0.1	0.3	0.9

*D = Diet and F = Feeding Level, D*F = Diet and Feeding level interactions.

compared to the other diet (Table 2). Biomass gain at low P:E diet was 1195 g.compartment⁻¹ compared to 985 g.compartment⁻¹, which is > 20% increase. Both diet ($P \leq .05$) and feeding level ($P \leq .001$) influenced growth (biomass gain per compartment) but there was no interaction effect between them. Even in non-fed compartments there was a diet effect and fish growth with the low P:E diet was 134% higher than the high P:E diet. Survival was not different between diets ($P > .05$) and increased with increasing feeding level ($P \leq .001$). FCR ranged between 0.5 and 1.6 and increased with increasing feeding level ($P \leq .001$). Difference in FCR between diets at high feeding level was higher than that at low feeding level. SGR was also higher in low P:E diet and was significantly influenced by diet ($P \leq .05$) and feeding level ($P \leq .001$).

4.2. Body composition

Initial body composition for dry matter (DM), crude protein (CP), crude fat (CFat) and ash were respectively 274, 157, 54, and 39 g.kg⁻¹. There was no effect of diet and feeding level or their interaction on final fish body composition (Table 3).

4.3. Feed nitrogen (N) input and output

At compartment level, feed N input was higher with the high P:E diet in fed compartments, as was intended. However, the N gain in fish per compartment was higher ($P \leq .05$) with the low P:E diet. The N gain increased with increasing feeding level (Table 4). At compartment level N balance was also influenced by the diet ($P \leq .001$) and the feeding level ($P \leq .001$).

4.4. Food web

There was no diet effect on the measured parameters of the food web (Table 5 and Supplementary Table 1). Abundance and diversity of phytoplankton, and total count of bacteria in soil and water column changed ($P \leq .001$) over time (Supplementary Table 1). Zooplankton diversity was influenced by feeding level ($P \leq .05$) and sampling time ($P \leq .001$). There was a trend towards significant effect of feeding level ($P \leq .08$) on abundance of zooplankton and time ($P \leq .07$) on

Table 4

Effect of dietary protein to energy ratio (P:E) and feeding level on feed N input and gain in fish.

Per compartment		“Low P:E” diet									“High P:E” diet									P values		
		NF			LF			HF			NF			LF			HF			D	F	D*F
N Feed	g	NA			30	±	0.7	72	±	1.1	NA			40	±	0.7	91	±	1.2	0.00	0.00	0.00
N gain Fish	g	10	±	1.1	36	±	3.9	56	±	4.0	4.4	±	1.1	32	±	3.8	48	±	4.0	0.04	0.00	0.70
N Balance	g	−10	±	1.1	−5	±	3.8	15	±	4.7	−4.4	±	1.1	7	±	3.8	43	±	4.7	0.00	0.00	0.25

*NF=No Feeding, LF = Low Feeding and HF=High Feeding, D = Diet and F=Feeding Level, D*F = Diet and Feeding level interactions.

Table 5

Effect of dietary protein to energy ratio (P:E) on food web in the pond.

		“Low P:E” diet									“High P:E” diet									P values		
Abundance of Phytoplankton	ind.l ^{−1}				71,201.3	±					4302.082			83,619.81	±				5934.276		0.121	
Diversity of Phytoplankton	genus.l ^{−1}				13.25926	±					0.544835			13.01852	±				0.598638		0.772	
Abundance of Zooplankton	ind.l ^{−1}				13,453.7	±					878.8398			14,361.11	±				1849.772		0.667	
Diversity of Zooplankton	genus.l ^{−1}				6.777778	±					0.353699			6.62963	±				0.145344		0.707	
Abundance of Benthos	ind.m ^{−2}				1244.4	±					470.6			1508.3	±				633.4		0.432	
Diversity of Benthos	group.m ^{−2}				2.777778	±					0.289742			3.111111	±				0.400617		0.515	
Chlorophyll a	µg.L ^{−1}				0.010389	±					0.000945			0.011778	±				0.000834		0.296	
Total count of bacteria from water	CFU.100 ml ^{−1}				41,818.52	±					4029.02			44,981.48	±				5763.505		0.662	
Total count of bacteria from soil	CFU.100 ml ^{−1}				95,992.59	±					2349.662			95,881.48	±				4005.571		0.981	

chlorophyll *a*. Variation between samples taken at the same time within diet and feeding level, was very high, especially for abundance of phytoplankton, zooplankton and total counts of both soil and water bacteria (Supplementary Table 1).

4.5. Water quality

There was a clear effect of time ($P < .001$) for all measured water quality parameters (Table 6). Dissolved oxygen concentration in the morning decreased over time from 4.25 mg.l^{−1} at the beginning of the experiment to 3.6 mg.l^{−1} at end. On the other hand, NO₂, NO₃ and NH₄⁺ increased over time (Table 6). However, for all time points, the measured water quality parameters were within the optimal range for Nile tilapia. N accumulation with the high P:E diet was higher by arithmetic mean compared to the low P:E diet but the difference by diet was not significant (Table 6).

5. Discussion

5.1. Effect of dietary protein to energy ratio on growth and body composition

We investigated the effect of dietary P:E ratio on fish performance and on the food web contribution to fish growth. The low P:E ratio diet (14 g.MJ^{−1}) performed better than the high P:E ratio diet (19 g.MJ^{−1}). The P:E ratio of better performed diet (14 g.MJ^{−1}) is lower than the

recommended P:E ratios (18–23 g.MJ^{−1}) for Nile tilapia (El-Sayed and Teshima, 1992; Kaushik et al., 1995; NRC, 1993). The previously reported lowest P:E ratio was 16 g.MJ^{−1} (Haidar et al., 2018). However, in Haidar et al. (2018) recommended diet CP content was 382 g.kg^{−1} and gross energy content was 23.5 kJ.g^{−1} which in the current study is 36% less for CP and 19% less for gross energy. Reducing the gross amount of CP and energy content in the diet might have practical implication on economic benefits. The reduced inclusion of CP in the diet has been complemented by the natural food of the pond. Several studies on different fish species cultured in the pond supported this outcome (Asaduzzaman et al., 2010a, 2010b; Bombeo-Tuburan et al., 1993; Jackson et al., 2013; Porchas-cornejo et al., 2012; Pucher and Focken, 2017; Rahman et al., 2008).

The proximate composition of formulated feed affects fish body composition. By lowering the dietary P:E ratio (Haidar et al., 2018), the fish fat content was 157 g.kg^{−1} and by reducing the dietary CP content (Al Hafedh, 1999) the fish fat content was 116 g.kg^{−1}. These studies were done in absence of natural food. The low 5.2% fat content in fish at harvest in the current experiment suggests the fish obtained protein not only from the feed but also from natural foods present in the pond.

5.2. Contribution of food web to fish growth in a pond

To answer the above question we calculated the amount of N coming from food web of the pond. We could not measure digestibility of nutrients in this experiment as it was conducted in ponds. So, we

Table 6

Effect of dietary protein to energy ratio (P:E) and Sampling time on pond water quality.

		“Low P:E” diet									“High P:E” diet									P values		
		T1			T2			T3			T1			T2			T3			D	T	D*T
Dissolved oxygen	mg.l ^{−1}	4.25	±	0.01	3.66	±	0.01	3.6	±	0.01	4.2533	±	0.01	3.6517	±	0.01	3.6	±	0.01	0.72	0.00	0.75
pH		7.31	±	0.01	7.26	±	0.01	7.67	±	0.01	7.31	±	0.01	7.25	±	0.01	7.66	±	0.01	0.54	0.00	0.93
Temperature	°C	30.5	±	0	31.2	±	0.04	31.4	±	0	30.497	±	0.05	31.15	±	0.04	31.4	±	0.04	0.64	0.00	0.80
Salinity	ppt	4.87	±	0.0	3.0	±	0.0	2.0	±	0.0	4.9	±	0.0	3.0	±	0.0	2.0	±	0.0	0.68	0.00	0.68
TDS	mg.l ^{−1}	5056	±	7.1	4001	±	2.62	3337	±	5.9	1.0	±	5061	2.0	±	4003	3.0	±	3339	0.07	0.00	0.96
Transparency	inch	11.1	±	0.2	10.8	±	0.43	11.9	±	0.3	10.75	±	0.23	10.833	±	0.43	11.9	±	0.32	0.59	0.01	0.88
NO ₂	mg.l ^{−1}	0.03	±	0	1.7	±	0.21	3.75	±	0.2	0.025	±	0	1.9	±	0.21	4.27	±	0.18	0.07	0.00	0.30
NO ₃	mg.l ^{−1}	0	±	0	5	±	0	7.5	±	1	0	±	0	5	±	0	5.83	±	0.99	0.26	0.00	0.26
NH ₄ ⁺	mg.l ^{−1}	0.5	±	0	1	±	0	1	±	0	0.5	±	0	1	±	0	1	±	0	0.40	0.00	0.10

*T1, 2 & 3 = Sampling Day 31, 45 and 60, D = Diet, T = Sampling time, D*T = Diet and time interaction.

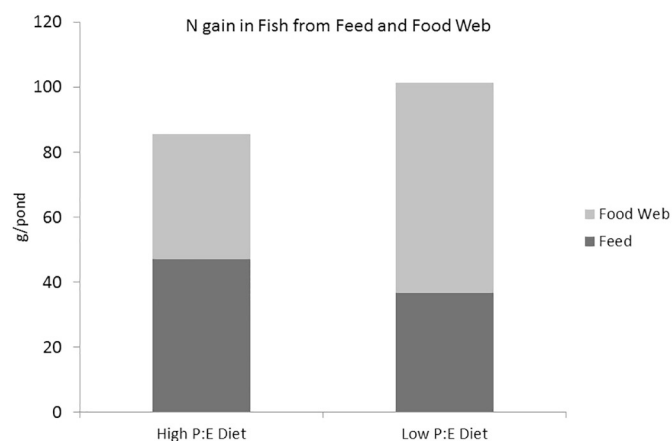


Fig. 2. Effect of dietary P:E ratio on the contribution of feed and food web to N gain in fish.

considered 90% protein digestibility (Azevedo et al., 2004; Kaushik et al., 1995) and 40% protein retention efficiency (RE) (Azevedo et al., 2004) for both diets and natural food. Under these assumptions, the contribution of natural food is 64% for the low P:E diet compared to 45% for the high P:E diet (Fig. 2). As the above estimation was based on logical assumption but not on actual result we did a sensitivity test by changing the ADC (80, 85, 90, and 95%) and RE (30, 35, 40, 45 and 50%) for both diet and natural food. The outcome largely remained same (Supplementary Fig. 1). These estimated contribution, underpin the importance of natural food web to fish production in semi-intensive ponds. Previous studies (Anderson et al., 1987; Burford et al., 2002, 2004; Cam and Mariotti, 1991; Porchas-cornejo et al., 2012) also confirmed that contribution of food web to fish growth can be between 40 and 68% in semi-intensive fed ponds based on the level of intensity and the type of species cultured. Findings of the current study remains within the previously reported results and we noticed that contribution of natural food decreased with an increasing feeding level (Fig. 3).

We further explored the composition of food web to understand the relative importance of different functional components, i.e. the pelagic and benthic, part of it. We considered that tilapia production realized in the non-fed compartment is based on the pelagic food web. As the water column was well mixed and turbulent due to aeration we assume the water column in each pond was quite uniform across compartments.

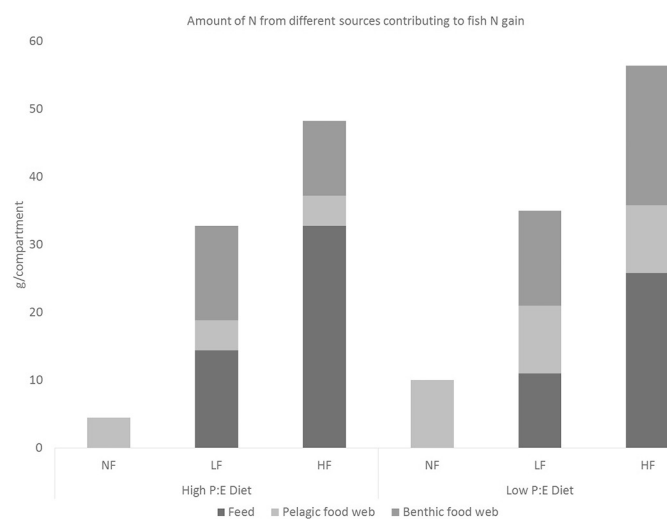


Fig. 3. Impact of dietary P:E ratio on contribution of fish N gain from feed, pelagic and benthic food web at different feeding levels (NF = no-feeding, LF = low feeding, and HF = high feeding).

When we measured the food web, we noticed presence of benthos in the non-fed compartments as well. However, we assumed that fish production in the non-fed compartment was gained only as an effect of the dissolved nutrients in the water column by the pelagic food web and their subsequent influence on other parts of the food web and fish.

The contribution of the benthic food web was estimated by deducting the contribution of feed and the pelagic food from the total achieved growth, in the remaining two fed compartments. By this calculation, contribution of the N from the pelagic food web was 30% vs 16% between low and high P:E ratio diet, and for benthic food web it was 34% vs 29% for the low and high P:E diet. At high feeding level contribution of the benthic food web was highest with the low P:E diet and lowest with the high P:E diet (Fig. 3). This is due to lack of adequate C in the waste of the pond fed by high P:E diet. So microorganisms cannot take up all the N in the waste. Wahab et al. (2003) reported a similar outcome of increased N accumulation and less growth in shrimp ponds. The striking thing here was that the effect of diet on the pelagic food web was different. In previous studies on pond food web enhancement by adding external carbon besides the diet prioritized the role of the bacterial and the benthic communities (Asaduzzaman et al., 2010b, 2010a, 2008; Avnimelech, 1999; Xu et al., 2018). The influence of lowering the dietary P:E ratio on both the pelagic and benthic food web in this study is probably due to making more carbon available for microbes to utilize available N in the pond. However, when we measured the food web we did not notice any difference between the diets (Supplementary Table 1). In extensive ponds, the food web follows a cyclic pattern of growth (Benincà et al., 2015) but in a well fed pond it should ideally remain stable unless controlled by predation. So, the likely cause of no difference in the food web was grazing pressure of the fish.

5.3. Varying Nitrogen retention efficiency with contrast dietary protein to energy level (or dietary protein level)

To estimate the effect of dietary P:E ratio on the fate of N in the system, we calculated total N input (g.pond^{-1}) including fertilizer and feed N, total N retained in the fish (g.fish^{-1}) and the non-retained N (g.pond^{-1}) in the pond. We observed N retention in fish (Fig. 4) is much higher from the low P:E diet (87%) compared to the high P:E diet (59%). The amount of N that is not retained in fish can cause higher total ammonia nitrogen and $\text{NO}_2\text{-N}$ concentrations at the water column (Hari et al., 2004) and in the sediment (Li and Lovell, 1992). In this study, we did not see the inorganic N (NO_2 , NO_3 and NH_4^+) accumulation in the water column. So, either this went to the pond bottom or returned to the atmosphere by denitrification. If the first case happens, high levels of ammonia could reduce the overall bottom productivity and fish yield (Wahab et al., 2003) unless there is adequate C in the

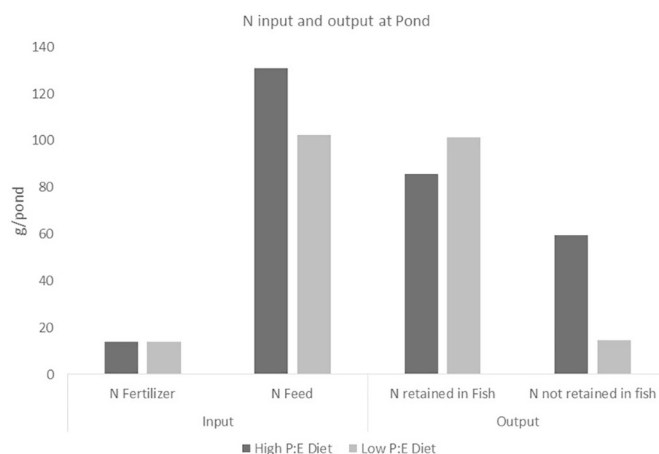


Fig. 4. Effect of dietary P:E ratio on N input, output and balance in the pond.

waste to maximize the use of additional N in the waste coming through high P:E diet. With the high P:E diet, the contribution of the benthic food web to fish production reduced with increasing feeding level. If the second case happens, it will return to atmosphere as N_2 , and if denitrification becomes carbon limited there is a risk of N_2O volatilization, contributing to global warming (Hu et al., 2012). In an ideal situation 98% of N input can be retained to fish biomass (Fan et al., 2015). In current study the RE of N with the low P:E diet (87%) is close to the reported best level (Fan et al., 2015).

6. Conclusion

Lowering the P:E ratio from 19 to 14 g.MJ⁻¹ in formulated feeds applied in ponds improved tilapia production. The P:E ratio of 14 g.MJ⁻¹ was much lower than the normally recommended P:E ratio of 18–23:E g.MJ⁻¹. Our analysis suggests that with the low P:E diet, fish consumed more natural foods to compensate for the lower nutrient input through the formulated feed. However, the higher contribution of natural foods to tilapia production was not shown by a higher abundance of plankton, benthos and soil and water microbiota. Hence, better parameters to explain natural food productivity and consumption by fish are needed. The latter will require additional research into the relation between the dietary P:E or C:N ratio, natural food productivity and the contribution of natural foods to total pond production.

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.aquaculture.2018.09.038>.

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